a bit clearer

C PROGRAMMING FROM SCRATCH · EPISODE 06

Keywords in C

Reserved Words and the 32 Classic Keywords

KEY POINTS

- Keywords, like int and return, belong to the language.
- You can't invent keywords or use one as a name.
- ANSI C / C89 defined exactly 32 classic keywords.
- int char; fails: char is a keyword, not a name.

01 What Is a Keyword?

▶ VIDEO 00:54

Keywords (reserved words) are words C reserves for its own grammar, each with a fixed job: `int` names the integer type, `return` leaves a function. The editor colours them blue.

FIGURE 6.1 **Two strict rules**

RULE 1

No new keywords

You cannot invent keywords of your own.

RULE 2

No keywords as names

When you name a variable or a function, you cannot use a keyword as that name.

02 The 32 Classic Keywords

▶ VIDEO 01:50

ANSI C / C89 defined **32 keywords**. No need to memorise them; they fall into a few groups.

GROUP	KEYWORDS
Data types and modifiers	`char` `int` `float` `double` `short` `long` `signed` `unsigned` `void`
Control flow (loops, branches)	`if` `else` `switch` `case` `default` `while` `for` `do` `break` `continue` `return`
Custom types	`struct` `union` `enum` `typedef`
Storage classes and qualifiers	`static` `extern` `const` `volatile`
Built-in operator	`sizeof`, which measures data size in bytes
The rest of the 32	`auto` `goto` `register`

03 Keywords in Later Standards

▶ VIDEO 04:00

STANDARD	NEW KEYWORDS
C99 (1999)	Five: `inline` `restrict` `_Bool` `_Complex` `_Imaginary`
C11	Keywords like `_Alignas` `_Atomic` `_Static_assert`
C23	Officially made `bool` `true` `false` standard keywords

`_Bool` is spelled that way so programs already using `bool` as a name didn't break. For beginners, the classic 32 are what matters.

04 Naming Collisions

▶ VIDEO 05:26

`int num;` builds cleanly: `int` is the type, `num` is our chosen name. Now change `num` to `char`:

keyword-identifier.c — C SOURCE

```c
// This file does not compile on purpose: char is a keyword, not a name.
#include <stdio.h>

int main()
{
    int char;
    return 0;
}
```

The build fails: the compiler expected a name but found the keyword `char`. MSVC reports:

MSVC · cl /W4 · code/mistakes/keyword-identifier.c

```
mistakes\keyword-identifier.c(6): error C2632: 'int' followed by 'char' is illegal
mistakes\keyword-identifier.c(6): warning C4091: ' ': ignored on left of 'int' when no variable
is declared
```

KEY IDEA

Keywords can never be names: otherwise `char` could mean your variable or the character type.

CODE 2 files on GitHub github.com/abitclearer/c-programming-from-scratch/tree/main/06-keywords-in-c